S2 Supplementary: Subsystem Theoretical Extensions

Context-Conditional Transition Geometry (CCTG), the Hippocampal--Entorhinal Landscape System (HELS), NDT-Q operational content sections, and candidate TRN substrates

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Relation to the main article

This supplement expands three appendix pointers from the main article (`NoeticDiffusionTheory.typ`, Appendix): **Subsystem extensions: CCTG and HELS**, **NDT-Q: operational content sections**, and **Candidate TRN substrates and CCTG**. None of these is a Stage-1 result. They are speculative or infrastructure-only extensions that must earn support through their own dedicated causal or reversal tests.

S2.0 Subsystem extensions: CCTG and HELS

Context-Conditional Transition Geometry (CCTG) is a speculative replacement for universal TRN–variance-gate language. It proposes a context- and circuit-dependent contribution to the drift field,

$$dX_t = \left[b_0(X_t, t) + u_{k(t)} K_{k,c}^{\mathrm{TRN}}(X_t)\right] dt + \Sigma_t dW_t. \tag{1}$$

Here $K_{k,c}^{\mathrm{TRN}}$ is a transition-field hypothesis, not an identified diffusion controller. CCTG predicts context-dependent changes in labeled representation transitions and must be evaluated against its own reversal and causal falsifiers. HELS is orthogonal: it proposes that hippocampal–entorhinal history and replay may update a context-conditioned landscape F_c. Neither proposal is a Stage-1 result; HELS currently has validated infrastructure but no validated mechanism result.

S2.1 NDT-Q: operational content sections

MNPS characterizes a global dynamical regime, not the identity of an experienced or represented content. NDT-Q therefore treats content as a separate, operational readout problem. Let a regime base be represented by $\pi_B : E \to B$, with compatible realizations $F_b = \pi_B^{-1}(b)$ and a content readout $\rho_b : F_b \to C$. A content section $s_{c(b)}$ satisfies $\rho_{b(s_{c(b)})} = c$.

This factorization supports a measurable episode profile $P_t = (c_t, \chi_t, S_t, R_t, V_t)$: content, closure/stability, self-anchoring, reachability, and valence. It is an explanatory mathematical architecture, not a claim that neural tissue literally instantiates a fibre bundle or that current animal decoding measures qualia. Current synthetic and population-code results are narrow operational evidence; failed centroid and transport tests remain failures under their frozen contracts. The constitutive reading remains philosophical and is not adjudicated by this framework.

First empirical precursor (content-generalization decoding, not phenomenology). In a public mouse visual-cortex electrophysiology dataset (illusory-contour paradigm), a frozen, genuinely blind pre-specification tested whether a decoder trained only on physically real, orientation-matched edge trials (never exposed to any illusory-contour trial) classifies held-out illusory-contour-evoked population activity as its correct orientation-matched real-edge counterpart. It does, at a session-level median rate of 0.743 (chance 0.5), beyond a temporal circular-shift null and a non-visual control-region check (which is at chance), and the effect requires the specific orientation-matched training axis rather than an arbitrary two-way split of real-edge trials. Gross pre-event locomotor state and pupil-indexed arousal do not account for the effect. This is an **internal validated result** for content-generalization decoding under this specific design—an order of analysis simpler than the content-fiber machinery above or a closure-signature transport measurement, since no trajectory geometry or transport tensor is computed. A companion, higher-order test of session- and orientation-general representational convergence (a frozen crossnobis-style centroid-distance margin) did **not** find a robust effect, and a companion transport-tensor proxy likewise found no convergence signal; both are rejected for their tested operationalizations, not merely inconclusive. No phenomenal report exists in mice, so none of this speaks to content **experience**—only to whether a specific population code generalizes across a stimulus manipulation.

S2.2 Candidate TRN substrates and CCTG

This section records a candidate TRN contribution and a phenomenological simulation mapping; it does not identify TRN as the biological implementation of NDT's variance schedule.

Evidence ceiling. The auditory PV/SOM TRN reanalysis archived as Zenodo 21774408 found a robust, dataset-bounded PV-linked contraction of MGB population noise dimensionality. It did not validate a full single-unit variance-gating mechanism, a stable SOM analogue, or identification of $r_{\text{true}}(t)/\sigma(t)$ with TRN firing. TRN anatomy and physiology motivate candidate circuit mechanisms (Crabtree, 2018; Halassa & Acsády, 2016), but that motivation is not evidence of the NDT mapping.

Candidate circuit sketch.

$$\text{Cortical state} \underset{\text{L6/L5}}{\rightarrow} \text{TRN activity} \underset{\text{inhibition}}{\rightarrow} \text{Thalamic relay} \underset{\text{gating}}{\rightarrow} \text{Candidate state-dependent gating} \quad (2)$$

1. **Candidate inputs:** Layer-6 corticothalamic neurons (carrying prediction errors and task demands), layer-5 projections, brainstem arousal systems, and hippocampal novelty signals converge on TRN sectors (Halassa & Kastner, 2017; Hádinger et al., 2022; Takata, 2019). These inputs can influence TRN firing patterns and thalamocortical state.
2. **Candidate consequences:** TRN output may change thalamocortical gain, timing, or population transition geometry (Lewis et al., 2015); their relation to a coarse NDT variance parameter is not yet identified.
3. **Observable signatures:** Thalamocortical dynamics can produce measurable rhythmic patterns—theta–gamma PAC, SO–spindle coupling, phase coherence—which we use to **estimate** a proxy $\hat{r}(t)$ via the $r(t)$ construction formula given in the main article's Rhythmic Variance Control section.

Critically, $\hat{r}(t)$ constructed from observables is a proxy for a functional schedule, not for a known TRN firing signal. Its convergent and predictive validity can be tested without claiming TRN causality. Closed-loop or optogenetic perturbation remains a future causal test, requiring mean-matched controls and simultaneous schedule and geometry readouts.

Simulation ansatz (candidate TRN → variance). A conductance-dependent mapping may be used in simulations:

$$\sigma(t) = \sigma_{\min} + (\sigma_0 - \sigma_{\min}) \exp(-\kappa g_{\text{TRN}}(t)), \quad \kappa > 0 \quad (3)$$

This equation is not an empirical law and has no identified κ in NDT units. It should be varied in simulations rather than reported as a physiological calibration.

Biological rationale and boundary. TRN inhibitory conductance, PV cells, channel dynamics, and electrical coupling can motivate simulations of state-dependent thalamocortical gating (Halassa & Acsády, 2016; Lewis et al., 2015; Li et al., 2024). They do not calibrate a coarse-grained NDT variance parameter. Any conductance range or variance ratio used in a simulation is therefore a sensitivity setting, not an empirical mapping or prediction.

Interneuron and channel specifics (mechanistic substrates). TRN neurons are predominantly GABAergic and parvalbumin-positive (PV+); PV TRN activity regulates spindle density and background spectral context (Thankachan et al., 2019). Sectoral heterogeneity (sensory vs. associative TRN) and CaV3.3 T-type Ca^(2+) channels support fast spindles (Fernandez et al., 2018), while electrical coupling synchronizes sectoral bursts at the network level (Li et al., 2024). On the cortical side, layer-6 corticothalamic neurons (L6 CT) provide state- and task-dependent drive to TRN, and layer-5 projections route content (Halassa & Kastner, 2017; Hádinger et al., 2022). Together, these elements are candidate substrates for inhibitory timing and population-state regulation. They do not yet instantiate an identified NDT variance gate.

CCTG successor hypothesis. A more conservative circuit extension treats TRN as potentially changing relative transitions in a context-dependent drift field instead of as a universal variance executor (Eq. 1). This CCTG framing is speculative; it must earn support through context-reversal and causal tests of labeled representation transitions.

Status ceiling

CCTG and the candidate TRN circuit are speculative; HELS has infrastructure but no mechanism result; NDT-Q is an operational content architecture, not a qualia measurement or an established neural fibre-bundle topology. None of these are required for, or should be read as strengthening, the Stage-1 Minimal-MNDM / functional RVC / MNPS claims in the main article.

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